US009841538B2

(12) **United States Patent**
Neil

(10) **Patent No.: US 9,841,538 B2**
(45) **Date of Patent: Dec. 12, 2017**

(54) **ANAMORPHIC OBJECTIVE ZOOM LENS**

(71) Applicant: **Cooke Optics Ltd.**, Thurmaston, Leicester (GB)

(72) Inventor: **Iain A. Neil**, Canobbio (IT)

(73) Assignee: **COOKE OPTICS LTD.**, Thurmaston, Leicester (GB)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 229 days.

(21) Appl. No.: **14/969,622**

(22) Filed: **Dec. 15, 2015**

(65) **Prior Publication Data**

US 2016/0103302 A1 Apr. 14, 2016

Related U.S. Application Data

(63) Continuation-in-part of application No. 14/683,297, filed on Apr. 10, 2015, now Pat. No. 9,239,449, (Continued)

(51) **Int. Cl.**

G02B 15/14	(2006.01)
G02B 13/18	(2006.01)
G02B 3/06	(2006.01)
G02B 15/177	(2006.01)
G02B 7/02	(2006.01)

(Continued)

(52) **U.S. Cl.**

CPC ***G02B 3/06*** (2013.01); ***G02B 7/028*** (2013.01); ***G02B 15/177*** (2013.01); ***G02B 27/0025*** (2013.01); *G02B 3/04* (2013.01); *G02B 13/002* (2013.01); *G02B 13/009* (2013.01); *G02B 13/0045* (2013.01); *G02B 13/08* (2013.01); *G02B 13/18* (2013.01); *G02B 15/14* (2013.01); *G02B 15/16* (2013.01); *G02B 15/20* (2013.01)

(58) **Field of Classification Search**

CPC G02B 15/177; G02B 13/009; G02B 13/18; G02B 15/14; G02B 13/0045; G02B 13/04; G02B 15/16; G02B 15/20; G02B 27/0025; G02B 13/002; G02B 13/08

USPC 359/664, 668, 670, 671, 738–740, 753, 359/784

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

3,990,785	A *	11/1976	Hirose	359/670
5,668,666	A *	9/1997	Suzuki	359/674

(Continued)

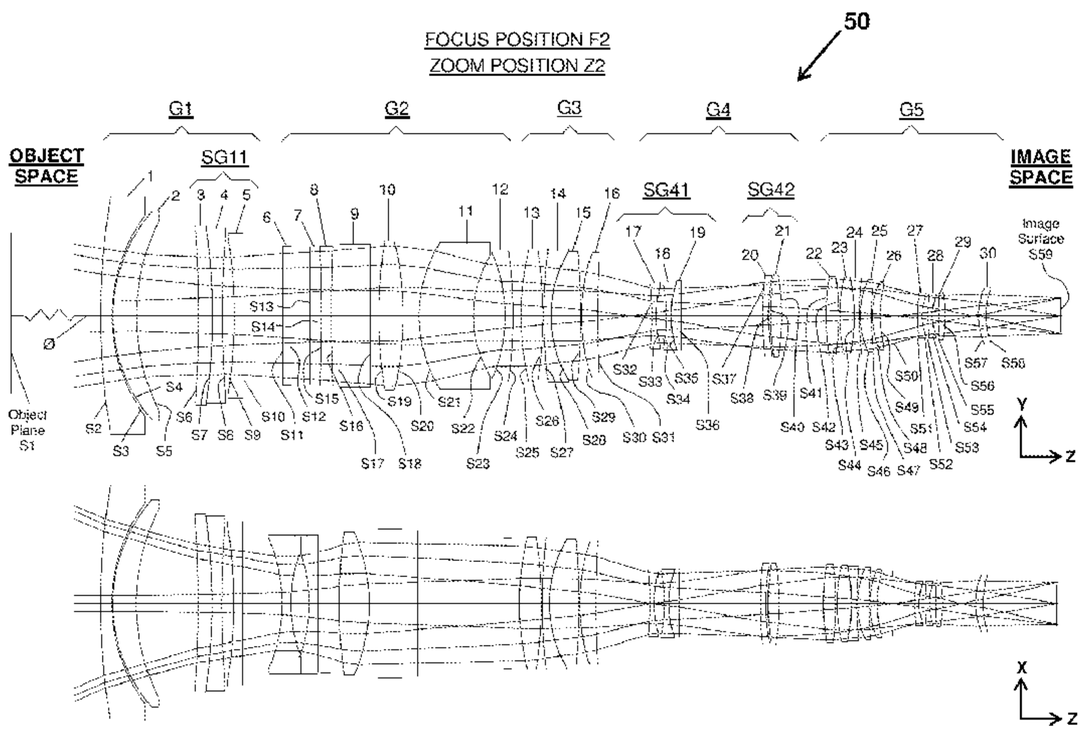
Primary Examiner — Evelyn A Lester

(74) *Attorney, Agent, or Firm* — Epstein Drangel LLP; Robert L. Epstein

(57) **ABSTRACT**

The anamorphic objective zoom lens includes, along an optical axis and in order from an object space to an image space: at least a negative (−) power spherical first lens group; an anamorphic second lens group, spherical third lens group preferably having positive (+) power, a variable power spherical fourth lens group and a positive (+) power spherical fifth lens group. The aperture stop is located before, after or preferably within the spherical fifth lens group. All spherical lens groups contain spherical and piano refractive optical surfaces. The anamorphic second lens group contains cylindrical and plano optical surfaces with at least one cylindrical surface oriented at substantially 90 degrees about at least one other cylindrical surface. The spherical first lens group may provide focusing. The variable power spherical fourth lens group provides zooming. The spherical third and fifth lens groups may provide compensation for thermal defocus.

35 Claims, 3 Drawing Sheets



Related U.S. Application Data

which is a continuation-in-part of application No. 14/218,064, filed on Mar. 18, 2014, now Pat. No. 9,341,827.

(60) Provisional application No. 61/808,343, filed on Apr. 4, 2013.

(51) **Int. Cl.**

G02B 27/00	(2006.01)
G02B 15/16	(2006.01)
G02B 13/00	(2006.01)
G02B 3/04	(2006.01)
G02B 15/20	(2006.01)
G02B 13/08	(2006.01)

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,745,298	A *	4/1998	Yahagi	359/683
6,122,111	A	9/2000	Neil et al.	
6,995,920	B2 *	2/2006	Nurishi	359/668
7,085,066	B2 *	8/2006	Neil	359/668
7,113,344	B2 *	9/2006	Nurishi et al.	359/668
7,123,421	B1	10/2006	Moskovich et al.	
8,662,679	B2 *	3/2014	Otani et al.	359/740
2014/0300973	A1	10/2014	Neil	

* cited by examiner

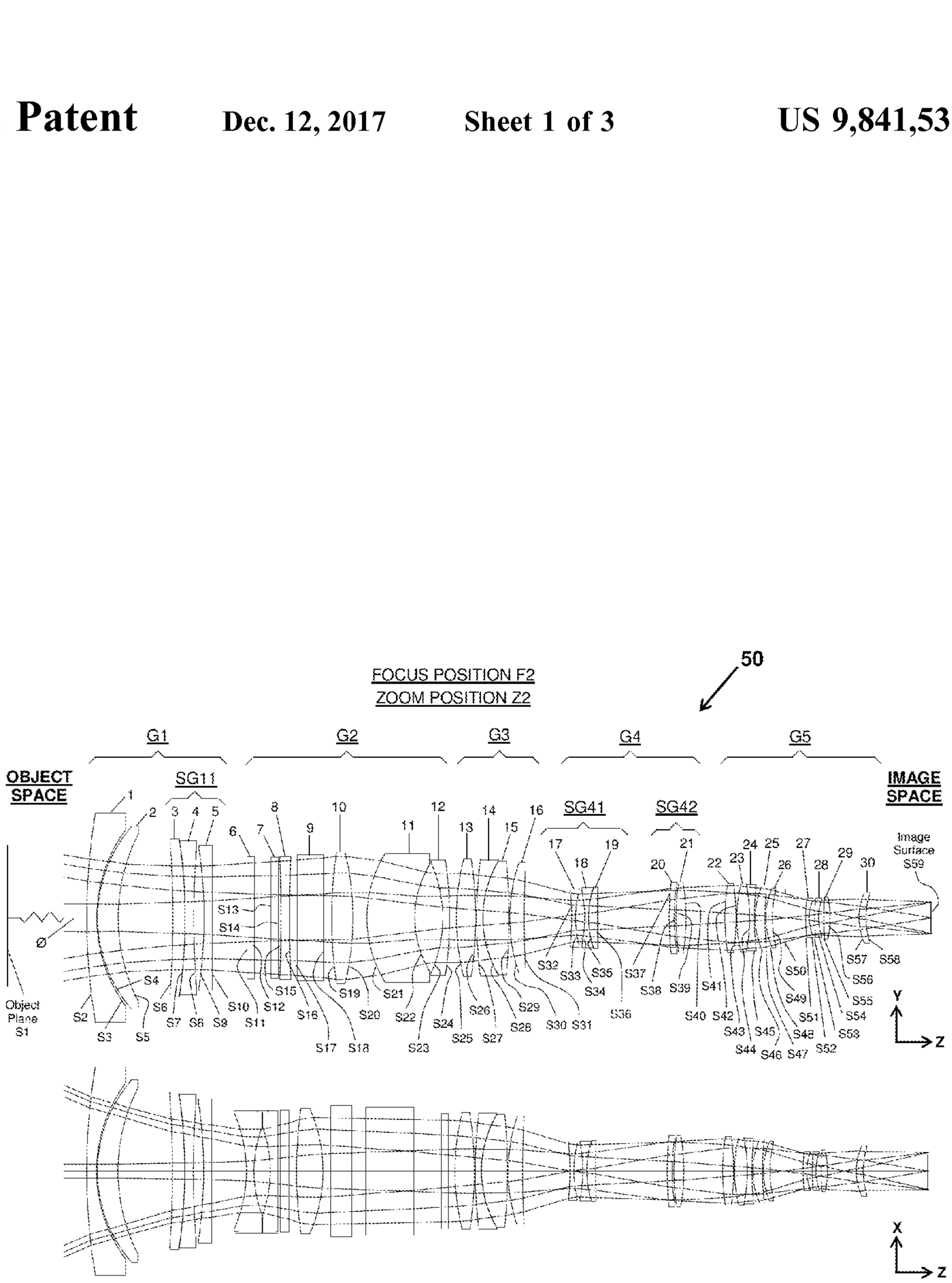

50
FOCUS POSITION F2
ZOOM POSITION Z2
G1
G2
G3
G4
G5
OBJECT SPACE
IMAGE SPACE
SG11
SG41
SG42
Object Plane S1
Image Surface S59
Ø
Y
Z
X
Z

FIG. 1

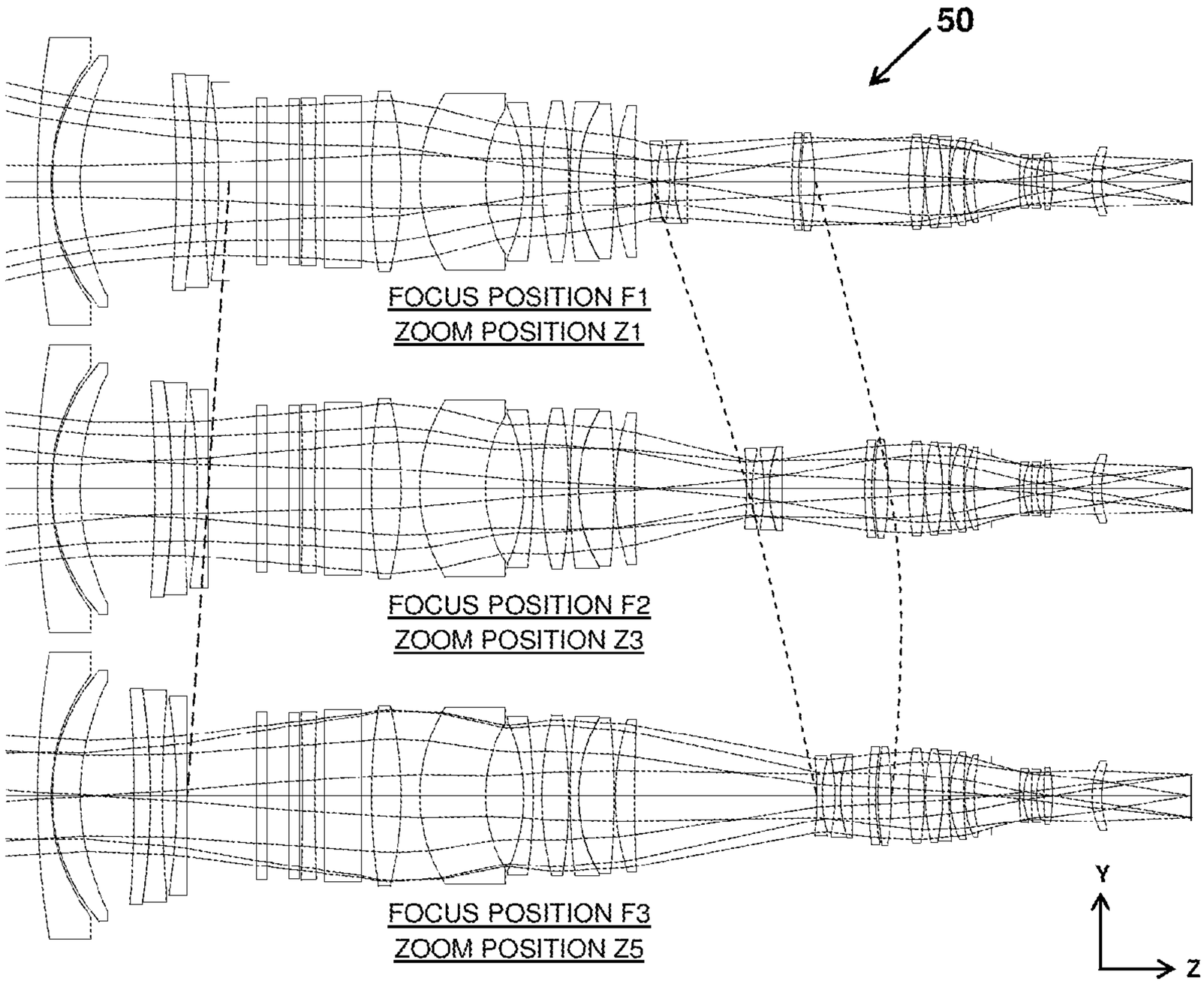

50
FOCUS POSITION F1
ZOOM POSITION Z1
FOCUS POSITION F2
ZOOM POSITION Z3
FOCUS POSITION F3
ZOOM POSITION Z5
Y
Z

FIG. 2

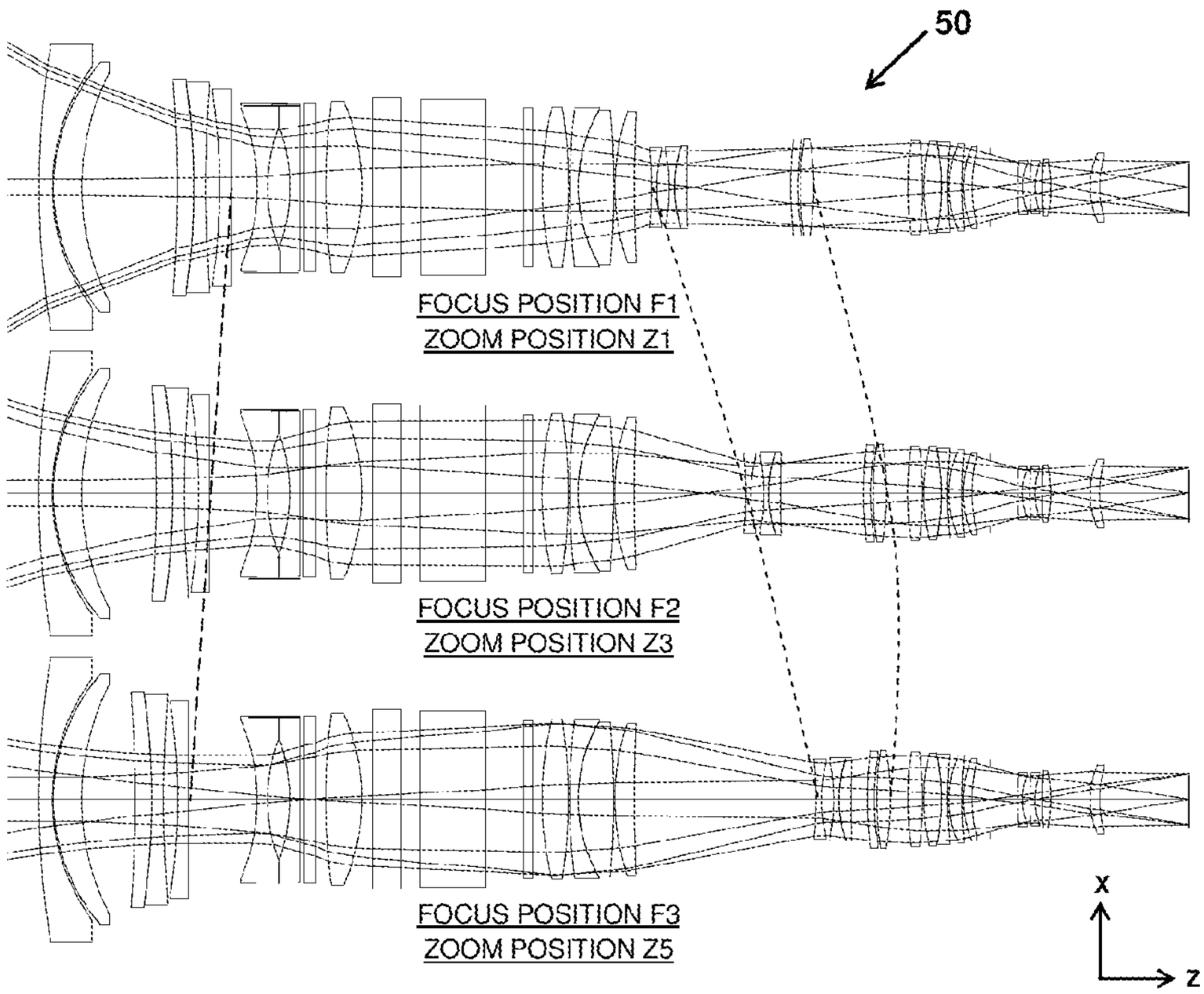

50
FOCUS POSITION F1
ZOOM POSITION Z1
FOCUS POSITION F2
ZOOM POSITION Z3
FOCUS POSITION F3
ZOOM POSITION Z5
X
Z

FIG.3

ANAMORPHIC OBJECTIVE ZOOM LENS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation in part of copending application Ser. No. 14/683,297 filed Apr. 10, 2015, which is a continuation is part of copending application Ser. No. 14/218,064, filed Mar. 18, 2014, which application claims the benefit under 37 CFR §119(e) of U.S. Provisional Application No. 61/808,343 filed Apr. 4, 2013, the contents of which are incorporated herein their entirety.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

Not Applicable

REFERENCE TO A "SEQUENCE LISTING", A TABLE, OR A COMPUTER PROGRAM LISTING APPENDIX SUBMITTED ON COMPACT DISC

Not Applicable

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to anamorphic objective zoom lenses, and more particularly to a range of different focal length anamorphic objective zoom lenses with zoom ratios that provide imaging over wide to narrow fields of view and provide traditional anamorphic imaging characteristics, and potentially compact diameter by having an arrangement of five lens groups with spherical, anamorphic and spherical powers, in which one of the spherical lens groups following the anamorphic lens group has variable power and an optical stop located inside the last spherical lens group.

2. Description of Prior Art Including Information Disclosed Under 37 CFR 1.97 and 1.98

Contemporary anamorphic objective zoom lenses normally have an optical axis and are commonly based on a rear anamorphic lens group or a front anamorphic lens group. Anamorphic objective zoom lenses having a rear anamorphic lens group are typically more commonplace than anamorphic objective zoom lenses having a front anamorphic lens group.

Anamorphic objective zoom lenses having a rear anamorphic lens group have a rear lens group with Y cylinder refractive optical surfaces and a front spherical lens group with an optical stop in the front spherical lens group in the form of a variable aperture diameter iris or diaphragm.

This anamorphic objective zoom lens arrangement produces images having spherical out of focus objects commonly referred to as the bokeh as compared to the oval or elliptically shaped out of focus objects produced by fixed focal length (commonly referred to as prime) anamorphic objective lenses. The oval or elliptically shaped bokeh of out of focus objects are desired by cinematographers because they produce a distinctive artistic look that is different from spherical objective lenses. Another common drawback with this anamorphic objective zoom lens arrangement is that the full aperture may be relatively slow as compared to that of anamorphic prime lenses.

Anamorphic objective zoom lenses having a front anamorphic lens group have a front lens group with X cylinder refractive optical surfaces and a rear spherical lens group with an optical stop in the rear spherical lens group in the form of a variable aperture diameter iris or diaphragm.

This anamorphic objective zoom lens arrangement produces images having oval or elliptically shaped out of focus objects commonly referred to as the bokeh which is desired by cinematographers for the reasons previously given however these lenses normally provide only small zoom ratios of 2× or 3×, where the smaller zoom ratio provides the widest field of view, and they tend be large in diameter with corresponding potentially higher weight and cost. They may also exhibit some breathing when focusing where the breathing is characterized by the field of view or focal length of the lens changing size as the lens is focused from distant to close objects or vice versa. In addition they may also produce a thermally induced focus shift with change of temperature where the focus shift is not constant over wide to narrow fields of view and the image is characterized by having residual astigmatism. Nevertheless the front anamorphic objective zoom lens arrangements produce images having numerous residual optical aberrations and characteristics most of which are desired by cinematographers because they produce an artistic look that is different from spherical objective lenses.

Many of the less desired residual optical aberrations and characteristics of these front and rear anamorphic objective zoom lens arrangements were accepted by cinematographers with film based cameras but with the advent and adoption of electronic sensor based digital cameras some of them have become less acceptable. In particular the amount of residual chromatic aberration has become less tolerable whereas some field curvature combined with some residual astigmatism is still acceptable.

As well as the oval or elliptically shaped bokeh another characteristic that is desired because of the distinctive artistic look produced is the depth of field being different in the vertical azimuth direction of the field versus the horizontal azimuth direction of the field. In the case of an anamorphic objective zoom lens that squeezes the horizontal field of view by substantially two times as compared to the vertical field of view, the depth of field in the horizontal azimuth direction of the field is substantially two times greater than the depth of field in the vertical azimuth direction of the field.

Improving the optical aberrations and characteristics of anamorphic objective zoom lenses of this arrangement may involve increasing optical surface shape complexity and hence manufacturing cost including adding aspherical and free-form shaped optical surfaces.

Thus, to address the artistic need of cinematographers and maximize the imaging potential of both film and digital cameras a compact diameter anamorphic objective zoom lens arrangement that provides a useful zoom range going from wide to narrow fields of view during zooming with a suitable blend of residual optical aberration correction and characteristics needs to be achieved.

Lenses of this type tend to be sensitive to changes in temperature and will go out of focus without some means of compensating for temperature shifts. Accordingly, there is a need for including some means of providing thermal compensation to maintain the focus over the temperature range in which the lens will operate.

BRIEF SUMMARY OF THE INVENTION

The present invention relates to an anamorphic objective zoom lens including, along an optical axis and in order from an object space to an image space, at least a negative (−)

power spherical first lens group: an anamorphic second lens group, a spherical third lens group preferably having a positive (+) power, a variable power spherical zoom fourth lens group and a positive (+) power spherical fifth lens group wherein an aperture stop is located before, after or preferably within the spherical fifth power lens group. The anamorphic second lens group has at least one cylindrical surface in a first direction and at least one other cylindrical surface in a substantially perpendicular direction to the first direction to enable a high degree of aberration correction over the whole image, whereby the residual longitudinal chromatic aberration and the residual lateral chromatic aberration are substantially reduced. The variable power spherical fourth lens group has at least two lens sub groups that are axially movable to provide zooming. The negative power spherical first lens group may provide focusing by movement of at least one of the lens elements contained therein and may exhibit low breathing in the focus range. Lens elements of the spherical third and fifth lens groups are independently axially movable by active or passive or active and passive means while preferably maintaining the overall length of the anamorphic objective zoom lens to provide thermal compensation of defocus which is consistently small and has little astigmatism through focusing and zooming. The lens is a complete assembly that forms a real image.

The anamorphic second lens group provides a squeeze of the field of view so that the focal lengths in the X and Y directions are different by a ratio of about two times which is typical for traditional anamorphic optical systems. The positive power spherical fifth lens group adjacent the image space delivers the radiation passing through the optical system on to the image sensor with nearly telecentric light paths and suitably high relative illumination, thereby increasing the efficiency of many electronic sensors.

The variable power spherical zoom fourth lens group which provides zooming uses at least two axially movable lens sub groups. The variable power spherical zoom fourth lens group is located between object space and the optical stop thus providing a substantially near constant aperture through zoom.

The lens elements of the spherical third and fifth lens groups which provide thermal compensation of defocus through focusing and zooming can be independently axially moved by active means employing motors or passive means utilizing materials exhibiting suitable thermal expansion coefficients in the mechanical support of the lens elements or a combination of both means. The thermal compensation of defocus by active, passive or combined means has the additional advantage of not requiring the operator of the lens to refocus the lens and the associated loss of focus scale calibration. The thermal compensation of defocus by passive means has the additional advantage of not requiring the complexity of a power source for motors.

In accordance with one aspect of the present invention, an anamorphic objective zoom lens is provided including, along an optical axis and in order from an object space to an image space, a negative (−) power spherical first lens group; an anamorphic second lens group, a spherical third lens group preferably having a positive (+) power, a variable power spherical zoom fourth lens group and a positive (+) power spherical fifth lens group and an aperture stop. The aperture stop is located in a position either before, after and within the spherical fifth lens group. At least one lens element of said spherical third lens group and at least one lens element of said spherical fifth lens group are independently axially adjustable to compensate for thermal defocus in image space for at least two focal lengths within the focal length range.

Preferably, the lens elements of the spherical third lens group are moveable together, the lens elements of the spherical fifth lens group are moveable together, and the spherical third lens group and the spherical fifth lens group are independently axially adjustable to compensate for thermal defocus in image space for at least two focal lengths within the focal length range.

Preferably, the aperture stop is located within the positive (+) power spherical fifth lens group.

The negative (−) power spherical first lens group is configured to provide focusing.

The anamorphic objective zoom lens has an optical axis. The first, second, third, fourth and fifth lens groups are situated along the optical axis. The anamorphic objective zoom lens is adapted for use between an object space and an image space. The optical axis extends between the object space and the image space.

The fifth lens group has at least one cylindrical surface in a first direction and at least one cylindrical surface in a direction substantially perpendicular to the first direction.

The fifth lens group has focal lengths in the X and Y directions which differ and together with the other lens group focal lengths produce focal lengths in X and Y directions which differ by a ratio of about two times.

The anamorphic objective zoom lens preferably has a focal length within the range of from at least 35 mm to 140 mm and preferably 40 mm to 125 mm in the Y direction.

The anamorphic objective zoom lens provides low residual chromatic aberration, a traditional oval bokeh shape, and different depths of field in the vertical and horizontal azimuth directions of the field.

The anamorphic objective zoom lens has a medium fast full aperture, moderately wide angle field of view to a moderately narrow angle field of view throughout its zoom lens range.

The lens groups of the anamorphic objective zoom lens are fabricated of lens elements made of glass.

The spherical lens groups include a lens element with a rotationally symmetrical surface shape about the optical axis.

The lens group with anamorphic powers includes a lens element with a non-rotationally symmetrical surface about the optical axis.

The anamorphic objective zoom lens preferably operates at an aperture of f/3.1 and over a waveband of 455-656 nm.

The anamorphic second lens group has seven cylindrically surfaced lens elements with eight Y cylinders, five X cylinders and one plano surface shapes.

The spherical first lens group includes five lens elements, three of which are axially moveable relative to the other.

The spherical third lens group includes four lens elements.

The spherical zoom fourth lens group includes five lens elements, three of which form a first zoom sub group and two of which form a second zoom sub group, both of which are axially movable.

The spherical fifth lens group includes nine lens elements.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF DRAWINGS

To these and to such other objects that may hereinafter appear, the present invention relates to an anamorphic objective zoom lens as described in detail in the following

specification and recited in the annexed claims, taken together with the accompanying drawings, in which like numerals refer to like parts and in which:

FIG. **1** is a lens plot in the YZ elevation (side view) and XZ elevation (plan view) on an optical axis Ø where the Y direction focal length is 51.00 mm and the X direction focal length is 26.21 mm. In the YZ elevation, three fields are shown at zero, top and bottom of the field of view. In the XZ elevation, three fields are shown at zero and both sides of the field of view. In the YZ elevation and in the XZ elevation diagrams an intermediate focus distance arrangement is shown;

FIG. **2** is a lens plot in the YZ elevation (side view) on an optical axis Ø where the Y direction focal lengths are 40.01 mm, 67.98 mm and 125.01 mm with three fields shown at zero, top and bottom of the field of view and the top to bottom diagrams showing far, intermediate and close focus distance arrangements; and

FIG. **3** is a lens plot in the XZ elevation (plan view) on an optical axis Ø where the X direction focal length are 20.57 mm, 34.94 mm and 64.27 mm with three fields shown at zero and both sides of the field of view and the top to bottom diagrams showing far, intermediate and close focus distance arrangements.

DETAILED DESCRIPTION OF THE INVENTION

The invention relates to anamorphic objective zoom lenses, and in particular to a range of different focal length anamorphic objective zoom lenses covering at least a focal length range from 35 mm to 140 mm and preferably 40 mm to 125 mm in the Y direction and providing low residual chromatic aberration, a traditional oval bokeh shape and different depths of field in the vertical and horizontal azimuth directions of the field.

The term "lens group" as used in connection with the anamorphic objective zoom lens disclosed herein means one or more individual lens elements. Also, the terms "optical stop" and "stop" are equivalent terms that can be used interchangeably.

The example provided herein is a preferred embodiment of the invention in which the first (front) lens group is negatively powered, the third lens group is positively powered, the fourth lens group is variable powered and the last (rear) lens group is positively powered. Those lens groups are paired with an anamorphic second lens group to work in unison and match the preferred optical interface characteristics of sensors, where near telecentric radiation beams approach the sensor.

The preferred embodiment discussed below is a medium fast full aperture moderately wide angle field of view to moderately narrow angle field of view anamorphic objective lens of the zoom.

In the preferred embodiment, all of the lens elements are made from glasses. The lens element optical surface shapes in the first, third, fourth and fifth spherical lens groups are all rotationally symmetrical about the optical axis such as spherical and in the anamorphic second lens group at least one lens element surface shape is non-rotationally symmetrical about the optical axis such as cylindrical.

In addition to providing these kinds of features and others like low breathing and (near) telecentric radiation output at the sensor, the preferred embodiment is capable of achieving suitable levels of various performance including image quality resolution and contrast (usually measured as MTF), high relative illumination for low shading and efficient optical throughput at the sensor via near telecentric radiation output at the sensor, which telecentric radiation output is less than 10 degrees.

The preferred embodiment of the present invention will now be described by way of a design example with accompanying figures and tables. Referring first to FIG. **1**, each lens element is identified by a numeral from **1** through **30** and the general configuration of each lens element is depicted, but the actual radius of each lens surface is set forth below in a table. The lens surfaces are identified by the letter "S" followed by a numeral from S**1** through S**58**.

Each lens element has its opposite surfaces identified by a separate but consecutive surface number as, for example, lens element **1** has lens surfaces S**2** and S**3**, lens element **11** has lens surfaces S**21** and S**22** and so forth, as shown in FIG. **1**, except that for doublet lens element pairs **3**, **4** and **14**, **15** and **18**, **19** and **23**, **24** the coincident facing lens surfaces are given a single surface number. For example, doublet lens element pair **3**, **4** is comprised of lens element **3** having a front lens surface S**6** and a rear lens surface S**7** and lens element **4** having a front lens surface S**7** (coincidental) and a rear lens surface S**8**. The location of the object to be photographed, particularly as it relates to focus distance, is identified by a vertical line and numeral "S**1**" on the optical axis, and the real image surface is identified by the numeral S**59**. All of the spherical lens surfaces have a finite radius of curvature except S**10** which is piano and all the cylindrically surfaced lens elements have a finite radius of curvature in X or Y directions except for S**14** which is piano.

Before describing the detailed characteristics of the lens elements, a broad description of the lens groups and their axial positions and movement will be given for the anamorphic objective zoom lens system of the present invention, generally designated **50**. Beginning from the end facing the object S**1** to be photographed. i.e. the left end in FIG. **1**, the spherical first lens group G**1** comprises lens elements **1** and **2** and a lens sub group SG**11** comprised of lens elements **3**, **4** and **5**. The anamorphic second lens group G**2** comprises lens elements **6**, **7**, **8**, **9**, **10**, **11** and **12**. The spherical third lens group G**3** includes lens elements **13**, **14**, **15** and **16**. The variable power spherical zoom fourth lens group G**4** includes, from left to right in FIG. **1**, lens sub group SG**41** including lens elements **17**, **18**, and **19**, and lens sub group SG**42** including lens elements **20** and **21**. The spherical fifth lens group G**5** closest to the image space includes lens elements **22**, **23**, **24**, **25**, **26**, **27**, **28**, **29** and **30**.

The images of FIG. **2** illustrate in the Y direction the movement of lens sub group G**11** in both directions along the optical axis for focusing at three focus positions and the movement of lens sub groups SG**41** and SG**42** in both directions along the optical axis for zooming at three positions.

The images of FIG. **3** illustrate in the X direction the movement of lens sub group G**11** in both directions along the optical axis for focusing at three focus positions and the movement of lens sub groups SG**41** and SG**42** in both directions along the optical axis for zooming at three positions.

In FIGS. **2** and **3** the auxiliary lens fifth group G**5** remains stationary and at a fixed distance from the real image surface S**59**.

While only the lens elements are physically shown in FIG. **1**, it is to be understood that conventional mechanical devices and mechanisms are provided for supporting the lens elements and for causing axial movement of the movable lens groups in a conventional lens housing or barrel.

The Optical Prescription data for the above described anamorphic zoom lens system **50** is set forth below in TABLE 1, which is extracted from data produced by CODE V® optical design software that is commercially available from Synopsis Optical Research Associates, Inc. Pasadena, Calif., U.S.A., which was also used for producing the optical diagrams. All of the data in TABLE 1 is given at a temperature of 25° C. (77° F.) and standard atmospheric pressure (760 mm Hg).

Throughout this specification, including the Tables, all measurements are in millimeters (mm) or as otherwise shown. In TABLE 1, the first column "ITEM" identifies each optical element and each location, i.e. object plane, etc., with the same numeral or label as used in FIG. **1**. The second and third columns identify the "Group" and "Subgroup", respectively, to which that optical element (lens) belongs with the same numerals used in FIG. **1**. The fourth column "Surface" is a list of the surface numbers and the fifth column "Shape" is the surface shape.

The sixth and seventh columns "Focus Position" and "Zoom Position", respectively, identify the typical focus positions of the spherical lens group and the typical positions of the lens elements in the variable power spherical zoom fourth lens group wherein there are changes in the distance (separation) between some of the surfaces listed in the "Separation" column which is the axial distance between that surface (fourth column) and the next surface, for example, the distance between surface S**2** to surface S**3** is 5.579 mm.

The columns headed by the legends "Y Radius of Curvature" and "X Radius of Curvature" list the optical surface radius of curvature for each surface in the Y and X plane, respectively, with a minus sign (−) meaning the center of the radius of curvature is to the left of the surface, as viewed in FIG. **1**, and "Flat" meaning an optically flat surface described as "Plano" in column **5**.

The next four columns, **11** to **14**, of TABLE 1 relate to the "Material" between that surface and the next surface to the right in FIG. **1**, with the eleventh column "Type" indicating whether there is a lens (Glass) or empty space (Air) between those two surfaces. All of the lenses are glass and the column titled "Code" identifies the optical glass. The column marked "Supplier" identifies the source of the lens and the column marked "Name" lists the Supplier's identification for each glass type, but it is to be understood that any equivalent or adequate glass may be used.

The last column of TABLE 1 headed "Aperture Half Diameter" provides the maximum aperture half diameter for each surface through which the light rays pass.

The novel configuration of having a negatively powered spherical first lens group, an anamorphic second lens group followed by a spherical third lens group preferably with a positive power, a variable power spherical fourth lens group and a positively powered spherical fifth lens group containing an optical stop may produce some residual distortion, astigmatism and field curvature aberrations but those aberrations to a tolerable extent contribute to the anamorphic look as desired by many cinematographers. In addition, a balanced blend of the afore-described lens characteristics may aid in cost reduction of manufacture. With the advent and adoption of digital cameras employing electronic sensors a large back focal length which was once required for film cameras having a reflex mirror may be less necessary but is still provided for in the novel anamorphic objective zoom lens.

The preferred embodiment operates at an aperture of f/3.1 and over a waveband of 455-656 nm and this waveband is what was used in the MTF Table 3. A faster or slower aperture may be required and an extended waveband may be required. The aperture may be increased or reduced and the waveband expanded and the optical designs re-optimized to maximize image quality over such apertures and wavebands without departing from the invention. Also, during such re-optimization alternate glass types may be used without departing from the spirit and scope of the disclosure. Furthermore, more complex optical surface shapes such as aspherical and free-form surfaces may be introduced for expanded performance but at the likely effect of increased manufacturing cost.

Referring to FIGS. **1-3**, those figures relate to the preferred embodiment in which the focal length in the Y directions are 40.01 mm, 67.98 mm and 125.01 mm and in the X directions are 20.57 mm, 34.94 mm and 64.27 mm. The overall length is 477 mm from the first refractive surface vertex of the lens to the image surface vertex, the front diameter clear aperture is 114.00 mm, the back focal length from the rear refractive surface vertex to the image vertex is 37.17 mm and the close focus distance from the object to the image is 1165 mm. The focal lengths of the spherical first lens group are −116.86 mm, −118.72 mm and −120.67 mm for the far, intermediate and close focus distances. The focal lengths of the anamorphic second lens group are +306.23 mm in the Y direction and −284.96 mm in the X direction. The focal length of the spherical third lens group is 107.33 mm. The focal lengths of the spherical fourth lens group with zooming are −90.88 mm, −100.10 mm and −70.74 mm for the short, medium and long focal length positions. The focal length of the spherical fifth lens group is 86.52 mm. The focal lengths of the focusing sub group and the zooming sub groups respectively are −265.40 mm, −46.15 mm and 190.83 mm.

FIGS. **2** and **3** show the locus of axial movement of the focusing sub group with a long dashed curve and the locus of axial movements of the zooming sub groups with short dashed curves where the closest approach of the zooming sub group nearest image space to the adjacent stationary lens element lies between zoom positions three (Z**3**) and five (Z**5**), of which the axial airspace distance of closest approach is given approximately by the data for zoom position Z**4** in Table 1 which accompanies this specification.

The focal lengths of the seven anamorphic lens elements containing at least one cylindrical surface are in order from an object space to an image space −75.75 mm (in X direction), −150.63 mm (in X direction), −561.89 mm (in Y direction), 87.29 mm (in X direction), 141.59 mm (in Y direction), −3906.70 mm (in Y direction) and −230.05 mm (in Y direction). It is to be understood that the focal lengths of the seven anamorphic lens elements in the other X and Y directions are substantially large and hence have little optical power.

In the preferred embodiment the lens system includes a total of thirty lens elements with twenty two singlets and four doublets. The spherical first lens group contains five lens elements with three elements axially movable for focusing at different distances, the anamorphic second lens group contains seven cylindrically surfaced lens elements with eight Y cylinders, five X cylinders and one plano surface shape, the spherical second group contains four lens elements, the zoom fourth lens group contains five lens elements which form two axially movable sub groups for zooming with three lens elements and two lens elements. The optical stop lies within the spherical fifth lens group. The nominal image size is 8.91 mm vertical half height and 10.65 mm horizontal half width in image space. In this

example embodiment the telecentric radiation output is about 9.1 degrees at all three focus positions and over the zoom range.

The accompanying Optical Prescription Table 1 describes a select example of the preferred embodiment of the anamorphic objective zoom lens disclosed herein.

Table 2 which accompanies this specification contains focal length, anamorphic squeeze, illumination and breathing data of the preferred embodiment. In Table 2 it is shown that the anamorphic squeeze ratio is within a small range of about 1.95% to 2.03%. In Table 2 it is also shown that the relative illumination is above 30%, which is sufficiently high for low shading at the corner of the field of view when an anamorphic objective zoom lens is used in combination with an electronic sensor at the image plane, such as when the anamorphic objective zoom lens constitutes part of a digital camera. In Table 2 it is further shown that the focus breathing is consistently very small through focusing and zooming.

In Table 3 which accompanies the specification, the polychromatic MTF performance at a spatial frequency of 20 cycles/mm is shown for the example embodiment to be greater than 50% at all field positions at the given combination of far, intermediate and close focus distances and short, medium and long focal lengths at a temperature of 25° C. (77° F.) and standard atmospheric pressure (760 mm Hg).

In Table 4 which accompanies the specification, the polychromatic MTF performance at a spatial frequency of 20 cycles/mm is shown for the example embodiment to be greater than 50% at all field positions at the given combination of far, intermediate and close focus distances and short, medium and long focal lengths at a temperature of 40° C. (104° F.) and standard atmospheric pressure (760 mm Hg). The third and fifth lens groups are axially moved to compensate for thermal defocus. The movements are given by respectively changing the separations of surfaces S**24**, S**31**, S**50** and S**58** by −0.126 mm, +0.126 mm, +0.054 mm and −0.054 mm.

In Table 5 which accompanies the specification, the polychromatic MTF performance at a spatial frequency of 20 cycles/mm is shown for the example embodiment to be greater than 50% at all field positions at the given combination of far, intermediate and close focus distances and short, medium and long focal lengths at a temperature of 10° C. (50° F.) and standard atmospheric pressure (760 mm Hg). The third and fifth lens groups are axially moved to compensate for thermal defocus. The movements are given by respectively changing the separations of surfaces S**24**, S**31**, S**50** and S**58** by mm, +0.126 mm, −0.126 mm, −0.054 mm and +0.054 mm.

In Tables 4 and 5 the polychromatic MTF of the anamorphic objective zoom lens system **50** has been computed theoretically in accordance with accepted optical principles and programs such as CODE V® optical design software. The glass coefficients of refractive index and expansion with temperature were obtained from commonly available data publications produced by the vendor corporations Ohara and Schott. All optical surface separations between lens elements are thermally controlled by the expansion and contraction of metal spacers made of aluminum contacting the optical surfaces at their clear apertures which are two times the aperture half diameter values given in Table 1. The temperature range of the anamorphic objective zoom lens system **50**, as designed and as described herein, is 10° C. (50° F.) to 40° C. (104° F.) with a nominal design temperature of 25° C. (77° F.). However, this anamorphic objective zoom lens system will maintain a high level of performance over at least an extended range of −10° C. to 50° C., although some degradation of image quality may be observed at low temperatures.

The seven anamorphic lens elements with the cylindrical surfaces of the example embodiment additionally may each have two refractive surfaces which may be formed by X and Y cylindrical surfaces or Y and X cylindrical surfaces with the X and Y surfaces substantially perpendicular to one another. This arrangement may improve the imaging characteristics but likely at the effect of additional manufacturing cost.

Although the present invention has been fully described in connection with a preferred embodiment thereof with reference to the accompanying drawings and data tables, various changes and modifications could be made thereto, including smaller and larger zoom ranges, smaller and larger focal lengths, smaller and larger anamorphic squeeze ratios, smaller and larger full aperture f/numbers, smaller and larger image sizes, smaller and larger wavebands, etc. (e.g., 435 nm to 656 nm), as will be apparent to those skilled in the art. Such changes and modifications are to be understood as being included within the scope of the present invention as defined by the appended claims.

TABLE 1

Optical Prescription

Item	Group	Sub-Group	Surface No.	Surface Shape[1]	Focus Position	Zoom Position[2,3]	Separation (mm)	Y Radius of Curvature (mm)	X Radius of Curvature (mm)
Object Plane			S1	Plano	F1	All	999999.000	Flat	Flat
					F2	All	1965.000		
					F3	All	943.000		
1	G1		S2	Sph.	All	All	5.579	364.892	364.892
			S3	Sph.	All	All	0.450	84.772	84.772
2	G1		S4	Sph.	All	All	11.657	79.239	79.239
			S5	Sph.	F1	All	39.560	105.024	105.024
					F2	All	30.907		
					F3	All	22.175		
3	G1	SG11	S6	Sph.	All	All	6.899	−585.944	−585.944
4	G1	SG11	S7	Sph.	All	All	4.873	−303.627	−303.627
			S8	Sph.	All	All	5.832	495.133	495.133
5	G1	SG11	S9	Sph.	All	All	4.517	−261.296	−261.296
			S10	Plano	F1	All	11.045	Flat	Flat
					F2	All	19.697		
					F3	All	28.430		

TABLE 1-continued

Optical Prescription								
6	G2	S11	X Cyl.	All	All	4.337	Flat	−82.662
		S12	X Cyl.	All	All	9.184	Flat	70.673
7	G2	S13	X Cyl.	All	All	4.249	Flat	−75.084
		S14	Plano	All	All	1.385	Flat	Flat
8	G2	S15	Y Cyl.	All	All	4.959	−908.544	Flat
		S16	Y Cyl.	All	All	4.334	920.220	Flat
9	G2	S17	X Cyl.	All	All	14.948	Flat	263.078
		S18	X Cyl.	All	All	4.367	Flat	−88.994
10	G2	S19	Y Cyl.	All	All	11.981	246.798	Flat
		S20	Y Cyl.	All	All	7.962	−161.076	Flat
11	G2	S21	Y Cyl.	All	All	27.028	69.953	Flat
		S22	Y Cyl.	All	All	15.808	58.918	Flat
12	G2	S23	Y Cyl.	All	All	3.800	−76.352	Flat
		S24	Y Cyl.	All	All	3.889	−232.274	Flat
13	G3	S25	Sph.	All	All	10.900	137.891	137.891
		S26	Sph.	All	All	0.519	−214.419	−214.419
14	G3	S27	Sph.	All	All	3.550	259.088	259.088
15	G3	S28	Sph.	All	All	14.662	64.133	64.133
		S29	Sph.	All	All	0.500	−353.262	−353.262
16	G3	S30	Sph.	All	All	8.144	97.834	97.834
		S31	Sph.	All	Z1	7.030	635.356	635.356
				All	Z2	26.780		
				All	Z3	46.204		
				All	Z4	64.237		
				All	Z5	75.329		
17	G4	SG41	S32	Sph.	All	All	2.200	−159.777	−159.777
			S33	Sph.	All	All	5.133	58.576	58.576
18	G4	SG41	S34	Sph.	All	All	2.200	−77.157	−77.157
19	G4	SG41	S35	Sph.	All	All	5.066	53.293	53.293
			S36	Sph.	All	Z1	43.249	880.564	880.564
					All	Z2	40.464		
					All	Z3	34.324		
					All	Z4	21.994		
					All	Z5	6.735		
20	G4	SG42	S37	Sph.	All	All	2.800	180.566	180.566
			S38	Sph.	All	All	1.047	109.323	109.323
21	G4	SG42	S39	Sph.	All	All	5.665	126.917	126.917
			S40	Sph.	All	Z1	38.886	−118.086	−118.086
					All	Z2	21.921		
					All	Z3	8.637		
					All	Z4	2.935		
					All	Z5	7.102		
22	G5		S41	Sph.	All	All	6.264	119.224	119.224
			S42	Sph.	All	All	0.500	−142.072	−142.072
23	G5		S43	Sph.	All	All	7.260	81.255	81.255
24	G5		S44	Sph.	All	All	2.600	−93.115	−93.115
			S45	Sph.	All	All	0.500	156.198	156.198
25	G5		S46	Sph.	All	All	5.501	44.318	44.318
			S47	Sph.	All	All	0.500	160.913	160.913
26	G5		S48	Sph.	All	All	3.694	42.727	42.727
			S49	Sph.	All	All	7.499	61.123	61.123
Stop			S50	Plano	All	All	11.321	Flat	Flat
27	G5		S51	Sph.	All	All	2.000	112.584	112.584
			S52	Sph.	All	All	3.094	26.037	26.037
28	G5		S53	Sph.	All	All	2.137	108.389	108.389
			S54	Sph.	All	All	3.023	30.081	30.081
29	G5		S55	Sph.	All	All	3.457	168.931	168.931
			S56	Sph.	All	All	16.389	−63.194	−63.194
30	G5		S57	Sph.	All	All	3.904	34.718	34.718
			S58	Sph.	All	All	37.175	44.308	44.308
Image			S59	Y & X Cy	F1	Z1	463.940	1029.861	
Surface				Y & X Cy	F2	Z1	390.380	−2461.934	
				Y & X Cy	F3	Z1	337.004	−589.389	
				Y & X Cy	F1	Z2	350.000	728.306	
				Y & X Cy	F2	Z2	315.862	−22158.909	
				Y & X Cy	F3	Z2	300.000	−1266.833	
				Y & X Cy	F1	Z3	331.907	984.085	
				Y & X Cy	F2	Z3	320.000	Flat	
				Y & X Cy	F3	Z3	300.000	−2186.394	
				Y & X Cy	F1	Z4	437.714	−6078.194	
				Y & X Cy	F2	Z4	388.741	−903.184	
				Y & X Cy	F3	Z4	294.968	−696.366	
				Y & X Cy	F1	Z5	816.007	−2031.177	
				Y & X Cy	F2	Z5	Flat	Flat	
				Y & X Cy	F3	Z5	550.000	−1039.893	

TABLE 1-continued

Optical Prescription										
		Sub-	Surface		Focus	Material				Aperture[6] Half Diameter
Item	Group	Group	No.	Shape[1]	Position	Type	Code	Name[4]	Supplier[5]	(mm)
Object Plane			S1	Plano	F1 F2 F3	Air				
1	G1		S2	Sph.	All	Glass	755523	SYGH51	OHARA	57.00
			S3	Sph.	All	Air				50.11
2	G1		S4	Sph.	All	Glass	805254	SF6	SCHOTT	49.83
			S5	Sph.	F1 F2 F3	Air				47.52
3	G1	SG11	S6	Sph.	All	Glass	805254	SF6	SCHOTT	42.75
4	G1	SG11	S7	Sph.	All	Glass	497816	SFPL51	OHARA	41.81
			S8	Sph.	All	Air				39.59
5	G1	SG11	S9	Sph.	All	Glass	618634	SPHM52	OHARA	39.18
			S10	Plano	F1 F2 F3	Air				38.28
6	G2		S11	X Cyl.	All	Glass	497816	SFPL51	OHARA	32.83
			S12	X Cyl.	All	Air				32.43
7	G2		S13	X Cyl.	All	Glass	497816	SFPL51	OHARA	32.42
			S14	Plano	All	Air				32.65
8	G2		S15	Y Cyl.	All	Glass	805254	SF6	SCHOTT	32.70
			S16	Y Cyl.	All	Air				32.95
9	G2		S17	X Cyl.	All	Glass	773.5	SLAH66	OHARA	33.41
			S18	X Cyl.	All	Air				33.90
10	G2		S19	Y Cyl.	All	Glass	694.51	SLAL58	OHARA	35.18
			S20	Y Cyl.	All	Air				35.45
11	G2		S21	Y Cyl.	All	Glass	487702	SFSL5	OHARA	34.72
			S22	Y Cyl.	All	Air				30.15
12	G2		S23	Y Cyl.	All	Glass	497816	SFPL51	OHARA	30.98
			S24	Y Cyl.	All	Air				31.12
13	G3		S25	Sph.	All	Glass	439950	SFPL53	OHARA	31.53
			S26	Sph.	All	Air				31.55
14	G3		S27	Sph.	All	Glass	720347	SNBH8	OHARA	31.26
15	G3		S28	Sph.	All	Glass	497816	SFPL51	OHARA	30.45
			S29	Sph.	All	Air				30.34
16	G3		S30	Sph.	All	Glass	439950	SFPL53	OHARA	29.87
			S31	Sph.	All All All All All	Air				29.38
17	G4	SG41	S32	Sph.	All	Glass	717479	SLAM3	OHARA	15.48
			S33	Sph.	All	Air				15.01
18	G4	SG41	S34	Sph.	All	Glass	589612	SBAL35	OHARA	15.14
19	G4	SG41	S35	Sph.	All	Glass	805254	SF6	SCHOTT	15.90
			S36	Sph.	All All All All All	Air				16.09
20	G4	SG42	S37	Sph.	All	Glass	801350	SLAM66	OHARA	18.84
			S38	Sph.	All	Air				18.83
21	G4	SG42	S39	Sph.	All	Glass	497816	SFPL51	OHARA	18.95
			S40	Sph.	All All All All All	Air				19.06
22	G5		S41	Sph.	All	Glass	439950	SFPL53	OHARA	18.30
			S42	Sph.	All	Air				18.24
23	G5		S43	Sph.	All	Glass	497816	SFPL51	OHARA	17.95
24	G5		S44	Sph.	All	Glass	750353	SNI3H51	OHARA	17.49
			S45	Sph.	All	Air				17.08
25	G5		S46	Sph.	All	Glass	497816	SFPL51	OHARA	16.91
			S47	Sph.	All	Air				16.40
26	G5		S48	Sph.	All	Glass	548458	STIL1	OHARA	15.84
			S49	Sph.	All	Air				15.14
Stop			S50	Plano	All	Air				13.55
27	G5		S51	Sph.	All	Glass	487702	SFSL5	OHARA	10.20
			S52	Sph.	All	Air				9.93
28	G5		S53	Sph.	All	Glass	516641	SBSL7	OHARA	10.07
			S54	Sph.	All	Air				10.16

TABLE 1-continued

Optical Prescription									
29	G5	S55	Sph.	All	Glass	487702	SFSL5	OHARA	10.54
		S56	Sph.	All	Air				10.84
30	G5	S57	Sph.	All	Glass	805254	SF6	SCHOTT	13.24
		S58	Sph.	All	Air				12.90
Image Surface		S59	Y & X Cy	F1	Air				
			Y & X Cy	F2					
			Y & X Cy	F3					
			Y & X Cy	F1					
			Y & X Cy	F2					
			Y & X Cy	F3					
			Y & X Cy	F1					
			Y & X Cy	F2					
			Y & X Cy	F3					
			Y & X Cy	F1					
			Y & X Cy	F2					
			Y & X Cy	F3					
			Y & X Cy	F1					
			Y & X Cy	F2					
			Y & X Cy	F3					

Notes:-

[1]In the Surface Shape column the image surface is not flat to simulate equivalent curved object surfaces through focus distance positions. F1, F2 and F3 and zoom positions Z1, Z2, Z3, Z4 and Z5.

[2]Paraxial focal lengths in the Y-direction at F1 focus position for zoom positions Z1, Z2, Z3, Z4 and Z5 respectively are 40.01 mm, 51.00 mm, 67.98 mm, 95.45 mm and 125.01 mm.

[3]Paraxial focal lengths in the X-direction at F1 focus position for zoom positions Z1, Z2, Z3, Z4 and Z5 respectively are 20.57 mm, 26.21 mm, 34.94 mm, 49.06 mm and 64.27 mm.

[4]In the Material Name column the trade name of the lens material used is given.

[5]In the Material Supplier column a manufacturer name is given although there may be alternative manufacturers.

[6]The data given in the Aperture Half Diameter column is for circular apertures.

TABLE 2

Focal Length, Anamorphic Squeeze, Illumination and Breathing

Zoom Position	Focus Position	Paraxial Y Direction	Focal X Direction	Anamorphic Squeeze Ratio[1]	Relative Illumination[2] (%)	Breathing[3]
Z1	F1	40.01	20.57	1.946	30.1	0.0
	F2	40.16	20.29	1.980	30.9	+0.4
	F3	40.33	20.04	2.013	30.8	+0.8
Z3	F1	67.98	34.94	1.946	30.1	0.0
	F2	68.65	34.52	1.989	30.1	+1.0
	F3	69.35	34.15	2.031	30.1	+2.0
Z5	F1	125.01	64.27	1.945	30.6	0.0
	F2	123.15	63.09	1.952	30.6	−1.5
	F3	121.48	62.03	1.958	30.6	−2.8

[1]Based on paraxial focal length in Y direction divided by paraxial focal length in X direction.

[2]At maximum radial image distance from the optical axis which is at the corner of the image.

[3]Based on the percentage difference between the paraxial effective focal length in the Y-direction at focus position F1 and focus positions F2 and F3.

TABLE 3

MTF Performance Data

FIELD POSITION										
Image Height		Normalized Image Height								
X-direction (mm)	Y-direction (mm)	X-direction (mm)	Y-direction (mm)	FOCUS POSITIONS (F) AND ZOOM POSITIONS (Z) F1, Z1	F3, Z1	F2, Z3	F1, Z5	F2, Z5	F3, Z5	PERFORMANCE DATA Description
0	0	0	0	91.0 (R)	90.9 (R)	89.7 (R)	74.7 (R)	75.4 (R)	78.2 (R)	Polychromatic diffraction
(Axial)		(Axial)		92.2 (T)	92.1 (T)	92.6 (T)	64.2 (T)	72.1 (T)	73.1 (T)	MTF data (%)
0	8.91	0	1	87.1 (R)	89.8 (R)	89.5 (R)	82.2 (R)	59.8 (R)	79.8 (R)	at 20 cycles/mm at image
(Top of Field)		(Top of Field)		72.5 (T)	69.3 (T)	75.7 (T)	78.8 (T)	74.4 (T)	69.2 (T)	surfaces and at the following
10.65	0	1	0	63.5 (R)	67.1 (R)	64.5 (R)	57.6 (R)	54.2 (R)	65.8 (R)	wavelengths 656.3, 587.6,
(Side of Field)		(Side of Field)		85.9 (T)	88.3 (T)	82.6 (T)	53.4 (T)	65.7 (T)	80.7 (T)	546.1, 486.1 and 455.0

TABLE 3-continued

MTF Performance Data										
FIELD POSITION										
Image Height		Normalized Image Height								
X-direction (mm)	Y-direction (mm)	X-direction (mm)	Y-direction (mm)	FOCUS POSITIONS (F) AND ZOOM POSITIONS (Z) F1, Z1	F3, Z1	F2, Z3	F1, Z5	F2, Z5	F3, Z5	PERFORMANCE DATA Description
7.47	6.24	0.7	0.7	74.2 (R)	74.7 (R)	81.1 (R)	80.0 (R)	82.3 (R)	79.6 (R)	nanometers with respective
(70% Corner to Field)		(70% Corner of Field)		78.6 (T)	78.6 (T)	83.9 (T)	64.7 (T)	75.3 (T)	77.4 (T)	weightings 7, 8, 9, 7 and 4,
10.65	8.91	1	1	67.1 (R)	66.1 (R)	57.6 (R)	53.1 (R)	63.5 (R)	71.7 (R)	where (R) = radial and (T) =
(Corner of Field)		(Corner of Field)		75.3 (T)	74.8 (T)	78.1 (T)	54.0 (T)	62.1 (T)	62.9 (T)	tangential azimuths

TABLE 4

MTF Performance Data										
FIELD POSITION										
Image Height		Normalized Image Height								
X-direction (mm)	Y-direction (mm)	X-direction (mm)	Y-direction (mm)	FOCUS POSITIONS (F) AND ZOOM POSITIONS (Z) F1, Z1	F3, Z1	F2, Z3	F1, Z5	F2, Z5	F3, Z5	PERFORMANCE DATA Description
0	0	0	0	90.3 (R)	90.3 (R)	89.3 (R)	73.2 (R)	74.0 (R)	76.9 (R)	Polychromatic diffraction
(Axial)		(Axial)		91.5 (T)	91.8 (T)	93.2 (T)	64.6 (T)	74.9 (T)	74.1 (T)	MTF data (%)
0	8.91	0	1	87.9 (R)	89.6 (R)	89.8 (R)	82.7 (R)	61.2 (R)	82.1 (R)	at 20 cycles/mm at image
(Top of Field)		(Top of Field)		70.5 (T)	67.0 (T)	73.5 (T)	79.8 (T)	71.6 (T)	65.6 (T)	surfaces and at the following
10.65	0	1	0	62.1 (R)	65.6 (R)	64.5 (R)	57.1 (R)	52.2 (R)	63.3 (R)	wavelengths 656.3, 587.6,
(Side of Field)		(Side of Field)		86.9 (T)	88.5 (T)	85.1 (T)	54.1 (T)	70.4 (T)	82.2 (T)	546.1, 486.1 and 455.0
7.47	6.24	0.7	0.7	73.3 (R)	73.7 (R)	80.5 (R)	79.6 (R)	81.4 (R)	78.1 (R)	nanometers with respective
(70% Corner to Field)		(70% Corner of Field)		77.6 (T)	77.3 (T)	84.9 (T)	66.1 (T)	77.1 (T)	77.4 (T)	weightings 7, 8, 9, 7 and 4,
10.65	8.91	1	1	67.9 (R)	66.7 (R)	58.3 (R)	54.0 (R)	64.4 (R)	71.5 (R)	where (R) = radial and (T) =
(Corner of Field)		(Corner of Field)		74.3 (T)	73.7 (T)	77.4 (T)	53.4 (T)	612 (T)	61.3 (T)	tangential azimuths

TABLE 5

MTF Performance Data										
FIELD POSITION										
Image Height		Normalized Image Height								
X-direction (mm)	Y-direction (mm)	X-direction (mm)	Y-direction (mm)	FOCUS POSITIONS (F) AND ZOOM POSITIONS (Z) F1, Z1	F3, Z1	F2, Z3	F1, Z5	F2, Z5	F3, Z5	PERFORMANCE DATA Description
0	0	0	0	91.3 (R)	90.9 (R)	90.4 (R)	77.2 (R)	77.9 (R)	80.3 (R)	Polychromatic diffraction
(Axial)		(Axial)		92.9 (T)	92.0 (T)	92.2 (T)	62.5 (T)	68.3 (T)	70.4 (T)	MTF data (%)
0	8.91	0	1	85.5 (R)	89.6 (R)	88.8 (R)	79.7 (R)	55.7 (R)	87.6 (R)	at 20 cycles/mm at image
(Top of Field)		(Top of Field)		73.6 (T)	70.8 (T)	76.4 (T)	77.2 (T)	77.6 (T)	71.1 (T)	surfaces and at the following
10.65	0	1	0	64.7 (R)	68.3 (R)	64.0 (R)	58.1 (R)	56.8 (R)	68.5 (R)	wavelengths 656.3, 587.6,
(Side of Field)		(Side of Field)		84.1 (T)	87.7 (T)	80.3 (T)	51.6 (T)	70.9 (T)	77.6 (T)	546.1, 486.1 and 455.0
7.47	6.24	0.7	0.7	75.3 (R)	75.7 (R)	81.4 (R)	80.0 (R)	83.0 (R)	81.2 (R)	nanometers with respective
(70% Corner to Field)		(70% Corner of Field)		79.4 (T)	79.5 (T)	83.1 (T)	61.9 (T)	72.7 (T)	76.0 (T)	weightings 7, 8, 9, 7 and 4,
10.65	8.91	1	1	65.2 (R)	64.0 (R)	55.9 (R)	50.3 (R)	61.3 (R)	71.4 (R)	where (R) = radial and (T) =
(Corner of Field)		(Corner of Field)		75.1 (T)	75.1 (T)	78.3 (T)	52.7 (T)	61.4 (T)	62.2 (T)	tangential azimuths

I claim:

1. An anamorphic objective zoom lens comprising along an optical axis and in order from an object space to an image space: a negative (−) power spherical first lens group; an anamorphic second lens group; a spherical third lens group, a zoom fourth lens group, a positive (+) power spherical fifth lens group and an aperture stop; wherein said aperture stop is located in a position selected from one of the following locations: within said spherical fifth lens group, between said zoom fourth lens group and said positive (+) power spherical fifth lens group and between said positive (+) power spherical fifth lens group and the image space,

wherein at least one lens element of said spherical third lens group and at least one lens element of said spherical fifth lens group are independently axially adjustable to compensate for thermal defocus in image space for at least two focal lengths within the focal length range.

2. The anamorphic objective zoom lens group of claim **1** wherein said spherical third lens group is a positive (+) power spherical third lens group.

3. The anamorphic objective zoom lens of claim **1** wherein said aperture stop is located within said positive (+) power spherical fifth lens group.

4. The anamorphic objective zoom lens of claim **1** having a focal length within the range of from at least 35 mm to 140 mm.

5. The anamorphic objective zoom lens of claim **1** having a focal length within the range of 40 mm to 125 mm.

6. The anamorphic objective zoom lens group of claim **1** wherein said spherical third lens group is a negative (−) power spherical third lens group.

7. The allomorphic objective zoom lens of claim **1** wherein said anamorphic second lens group has at least one cylindrical surface in a first direction and at least one cylindrical surface in a direction substantially perpendicular to said first direction.

8. The anamorphic objective zoom lens of claim **1** wherein said anamorphic second lens group has focal lengths in perpendicular directions which differ by a ratio of about two times.

9. The anamorphic objective zoom lens of claim **1** wherein said lens groups are fabricated of lens elements made of glass.

10. The anamorphic objective zoom lens of claim **1** wherein said spherical lens groups each comprise a lens element with a rotationally symmetrical surface shape about said optical axis.

11. The anamorphic objective zoom lens of claim **1** wherein said anamorphic second lens group comprises a lens element with a non-rotationally symmetrical surface about said optical axis.

12. The anamorphic objective zoom lens of claim **1** wherein said anamorphic second lens group comprises seven cylindrically surfaced lens elements with eight Y cylinders, five X cylinders and one plano surface shapes.

13. The anamorphic objective zoom lens of claim **1** wherein said spherical first lens group comprises five lens elements, three of which are axially moveable relative to the other.

14. The allomorphic objective zoom lens of claim **1** wherein said spherical third lens group comprises four lens elements.

15. The anamorphic objective zoom lens of claim **1** wherein said spherical fifth lens group comprises nine lens elements.

16. The anamorphic objective zoom lens of claim **1** wherein said zoom fourth lens group comprises five lens elements.

17. The anamorphic objective zoom lens of claim **1** wherein said zoom fourth lens group comprises two lens sub groups, said sub lens groups having three lens elements and two lens elements, respectively, of which both lens sub groups are axially moveable.

18. The anamorphic objective zoom lens of claim **1** wherein said spherical third lens group comprises lens elements and said spherical fifth lens group comprises lens elements, wherein said lens elements of said spherical third lens group are moveable together, said lens elements of said spherical fifth lens group are moveable together, and said spherical third lens group and said spherical fifth lens group are independently axially adjustable to compensate for thermal defocus in image space for at least two focal lengths within the focal length range.

19. The anamorphic objective zoom lens of claim **1** wherein thermal defocus in image space is compensated in said focal length range.

20. The anamorphic objective zoom lens of claim **1** wherein said anamorphic objective zoom lens has a substantially constant overall length.

21. The anamorphic objective zoom lens of claim **1** having a focal length within the range of from at least 35 mm to 140 mm in.

22. The anamorphic objective zoom lens of claim **1** wherein said third lens group comprises an element and said spherical fifth lens group comprises an element, wherein said element of said spherical third lens group and said element of said spherical fifth lens group are moveable.

23. An anamorphic objective zoom lens comprising along an optical axis extending from an object space to an image space: a spherical lens group adjacent the object space; a spherical lens group adjacent the image space; an anamorphic lens group located between said spherical lens group adjacent the object space and said spherical lens group adjacent the image space; a zoom lens group; a further spherical lens group situated between the anamorphic lens group and the zoom lens group, and an aperture stop situated within one of the following locations: within said spherical lens group adjacent the image space, between said spherical lens group adjacent the image space and said zoom lens group and between said spherical lens group adjacent the image space and the image space wherein at least one lens element of said further spherical lens group and at least one lens element of said spherical lens group adjacent the image space are independently axially adjustable to compensate for thermal defocus in image space for at least two focal lengths within the focal length range.

24. The anamorphic objective zoom lens of claim **23** wherein said spherical lens group adjacent the object space has a negative (−) power and said further spherical lens group has a positive (+) power.

25. The anamorphic objective zoom lens of claim **23** wherein said further spherical lens group has a positive (+) power and said spherical lens group adjacent the image space has a positive (+) power.

26. The anamorphic objective zoom lens of claim **23** wherein said anamorphic lens group comprises at least one cylindrical surface in a first direction and at least one cylindrical surface in a direction substantially perpendicular to said first direction.

27. The anamorphic objective zoom lens of claim **23** wherein said anamorphic lens group comprises at least one cylindrical surface in an X direction and at least one cylindrical surface in a Y direction.

28. The anamorphic objective zoom lens of claim **23** wherein the anamorphic objective zoom lens has a substantially constant overall length.

29. The anamorphic objective zoom lens of claim **23** having a focal length within the range of from at least 35 mm to 140 mm.

30. The anamorphic objective zoom lens of claim **23** wherein said lens element of said further spherical lens group and said lens element of said spherical lens group adjacent said image space are moveable.

31. The anamorphic objective zoom lens of claim **23** wherein said further spherical lens group comprises a lens element and said spherical lens group comprises a lens

element, wherein said lens element of said further spherical lens group and said lens element of said spherical lens group adjacent said image space are moveable.

32. The anamorphic objective zoom lens of claim **23** wherein said further spherical lens wrong comprises a lens element and said spherical lens group comprises a lens element, wherein said lens elements of said further spherical lens group are moveable together, and said lens elements of said spherical fifth lens group are moveable together, and said further spherical lens group and said spherical fifth legs group are independently axially adjustable to compensate for thermal defocus in image space for at least two focal lengths within the focal length range.

33. The anamorphic objective zoom lens of claim **32** wherein thermal defocus in image space is compensated within said focal length range.

34. The anamorphic objective zoom lens of claim **32** wherein the anamorphic objective lens has a substantially constant overall length.

35. The anamorphic objective zoom lens of claim **32** having a focal length within the range of from at least 35 mm to 140 mm.

* * * * *

UNITED STATES PATENT AND TRADEMARK OFFICE

CERTIFICATE OF CORRECTION

PATENT NO. : 9,841,538 B2
APPLICATION NO. : 14/969622
DATED : December 12, 2017
INVENTOR(S) : Iain A. Neil

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

On the Title Page

Item (72) should read:
(72) Inventor: **Iain A. Neil**, Canobbio (TI) (CH)

Signed and Sealed this
Thirteenth Day of March, 2018

Andrei Iancu
Director of the United States Patent and Trademark Office